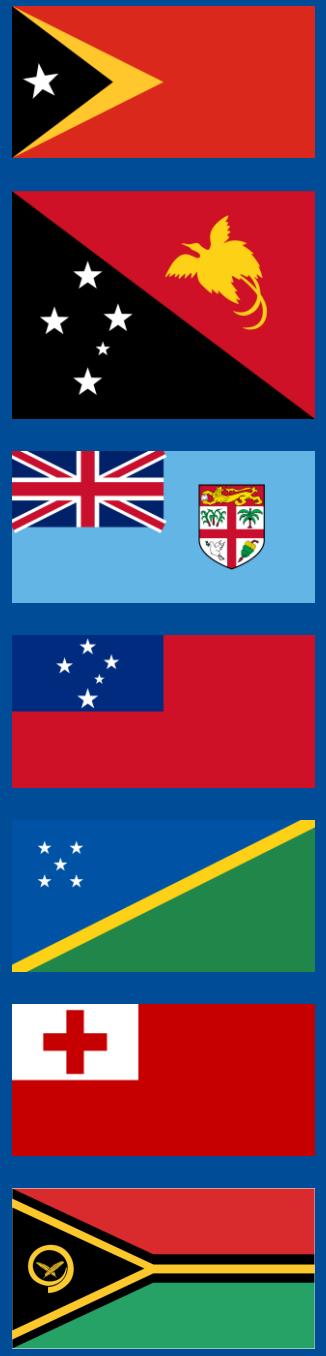


W-3 High-Frequency Indicators and Non-Traditional Data Workshop

OT26.07 Cohort Training in Forecasting and Nowcasting
September 21 – October 2, 2026

Jing Xie
Technical Advisor

This training material is the property of the International Monetary Fund (IMF) and is intended for use in IMF courses. Any reuse requires the permission of the IMF.

Workshop Objectives

Extract Non-Traditional Data for Target Countries

Google Trends

UNBigData

Data Science Campus

DEVELOPMENT DATA PARTNERSHIP

Apply Necessary Data Processing Procedures to Non-Traditional Data

- Fill-in Missing Data
- \+
- Remove Outliers
- \+
- Seasonal Adjustment
- \+
- Data Transformation

“Forecasting Private Consumption” from Torsten Schmidt and Simeon Vosen

Table 1

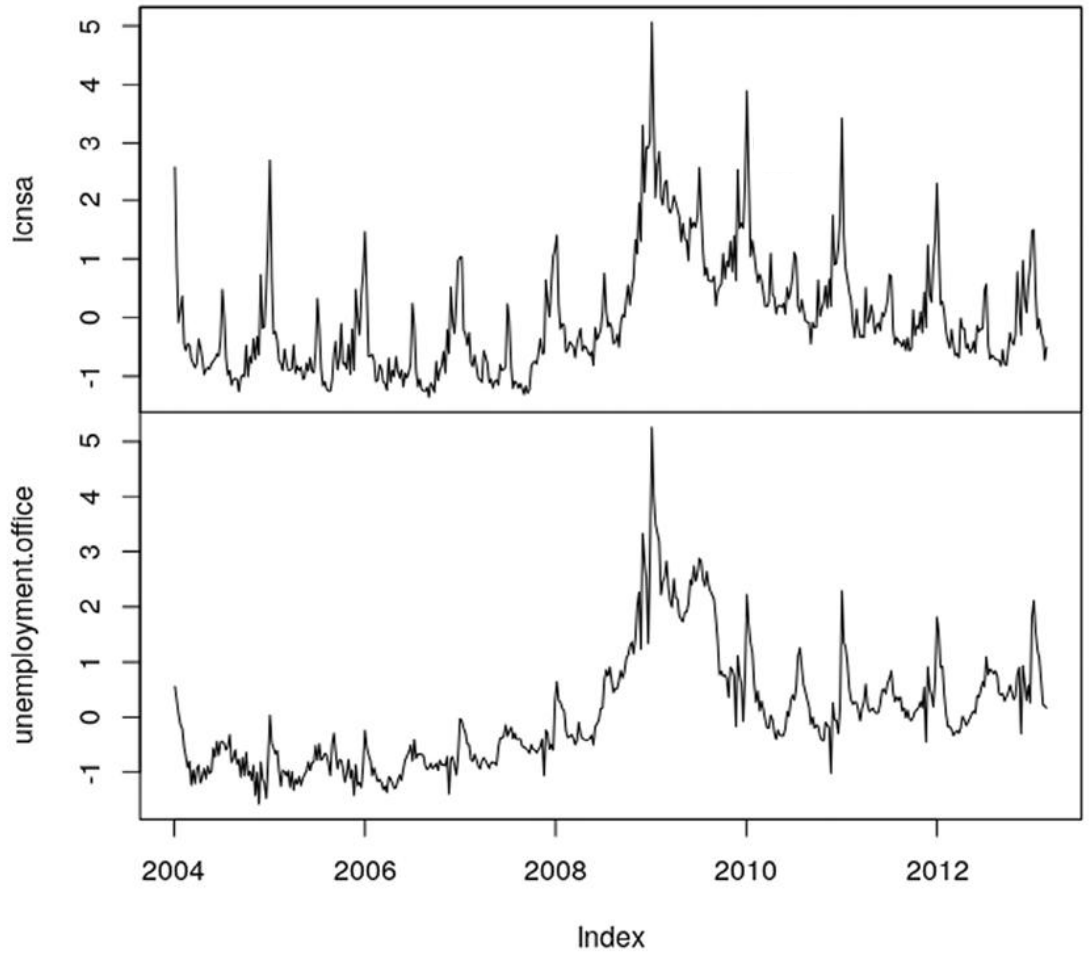
BEA classification of personal consumption expenditures (PCE) and matching Google categories

PCE by major type of product as classified by the national product and income accounts (NIPAs)	Google categories
Durable goods	
Motor vehicles and parts	Automotive, Auto Parts, Auto Financing, Auto Insurance, Vehicle Brands, Vehicle Shopping
Furnishings and durable household equipment	Computers & Electronics, Consumer Electronics, Home Appliances, Home Financing, Home Furnishings, Home & Garden, Home Improvement, Home Insurance, Homemaking and Interior Decoration, Interior Design
Recreational goods and vehicles	Book Retailers, Entertainment, Entertainment Industry, Movies, Video Games
Other durable goods	Book Retailers, Mobile & Wireless, Telecommunications

- This study introduces a new indictor for private consumption based on search query time series from Google Trend.
- The forecasting performance of the new indicator is assessed relative to the two most common survey-based indicators (the University of Michigan Consumer Sentiment Index and the Conference Board Consumer Confidence Index).
- The results show that in almost all conducted in-sample and out-of-sample forecasting experiments, the Google Indicators outperforms the survey-based indicators.

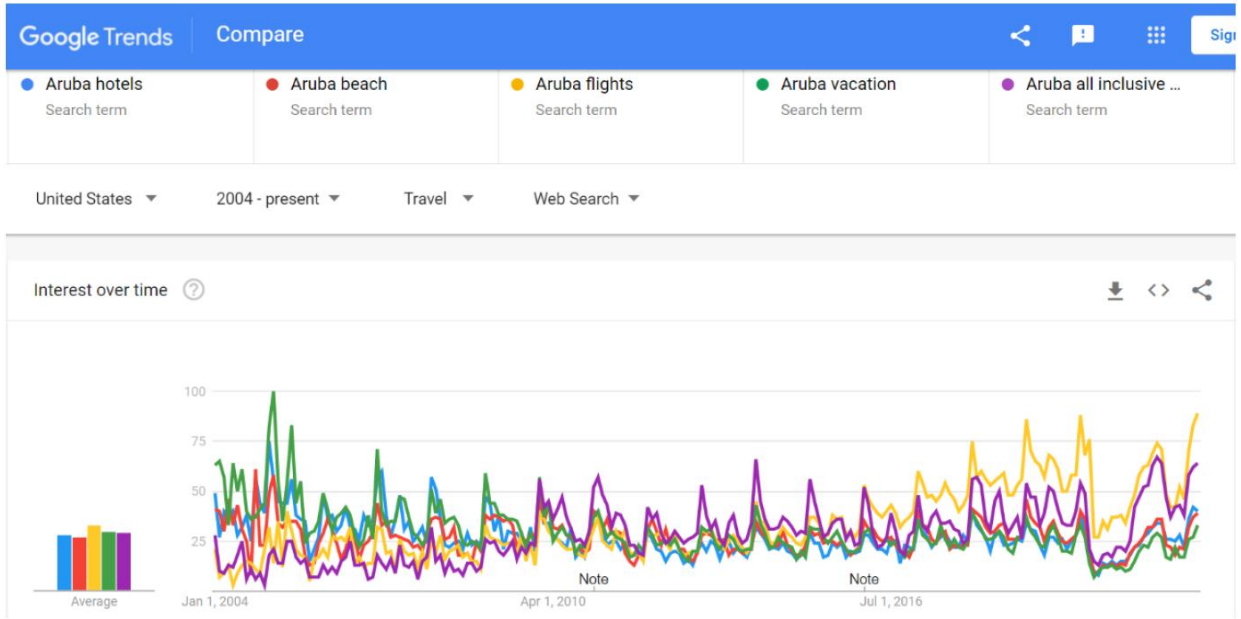
“Nowcasting with Google Trends” from Hal Varian

FIGURE 1
Initial Claims [Icnsa] from the Federal Reserve Economic Data (FRED) site and Searches for ‘Unemployment Office’ from Google Trends



- John Battelle once said, “Google is a database of intentions” (Battelle, 2005).
- The figure on the left illustrates the connection between the initial claims unemployment rate released by the Federal Reserve Economic Data (FRED) and Google Searches for “Unemployment Office”.
- The similarity between the two series is reflected in the Pearson correlation: 0.79.

“Modeling and Forecasting Monthly Tourism Arrivals to Aruba Since COVID-19 Pandemic” from Olga Bespalova



- This paper aims to enhance the accuracy of forecasting monthly tourism arrivals to Aruba, incorporating ARIMA-X models and various exogenous variables such as Google Trends and Flight capacity utilization.
- The study finds that search frequencies related to “Aruba all-inclusive hotels” provides valuable predictive information.
- Graphical analysis indicated that its dynamic were the closest among available Google Trends data in the US tourist arrivals to Aruba.

Step 1. Extract Data from Google Trend

Three Target Countries

We wish to extract **Google Trends** data relevant to Tourism for one target country:

Google Trends

https://trends.google.com/trends



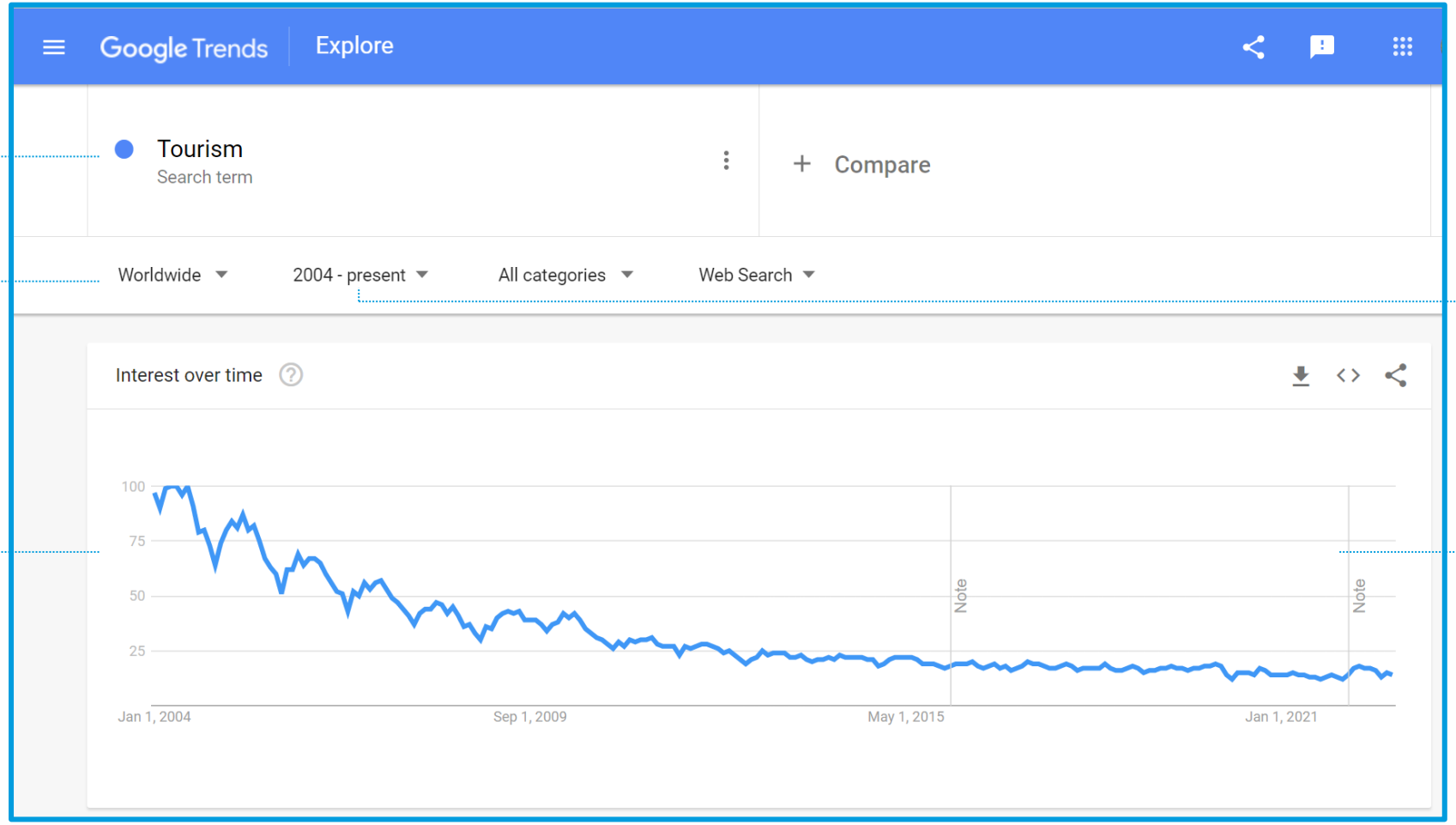
India

Key Words to Use:

- India Travel
- India Hotel
- Flight to India
- Taj Mahal

Step 1. Extract Data from Google Trend

Google Trends Interface



Keywords

Search in Specific Region

Google Trend Index (0~100)

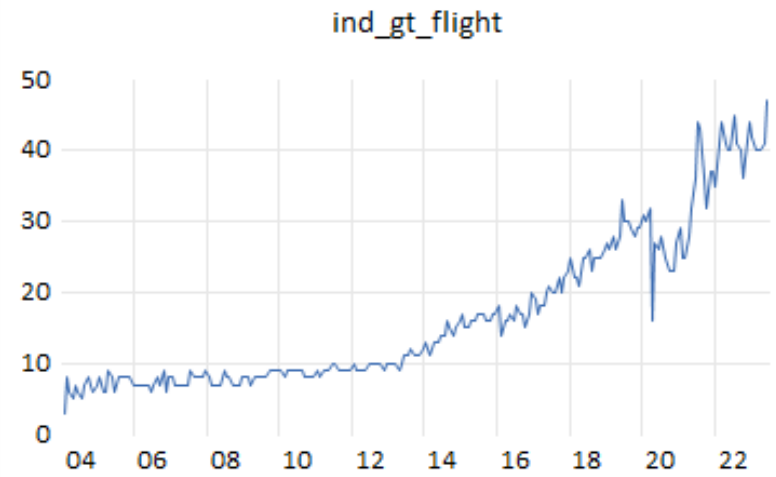
Time Coverages

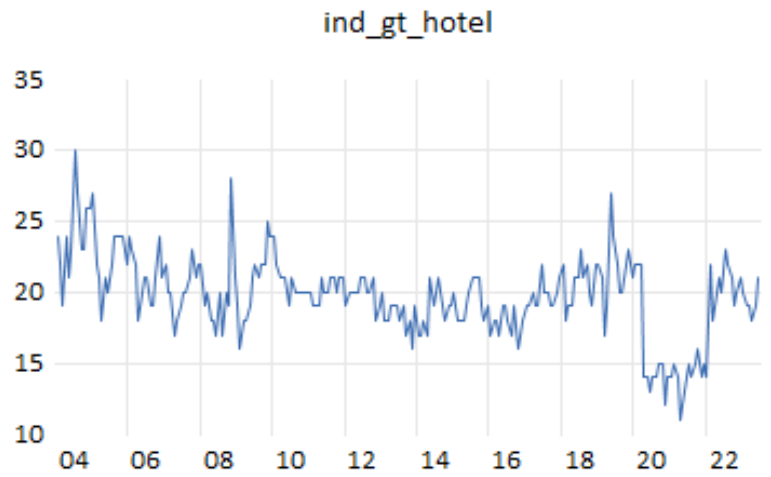
Interest Overtime Chart

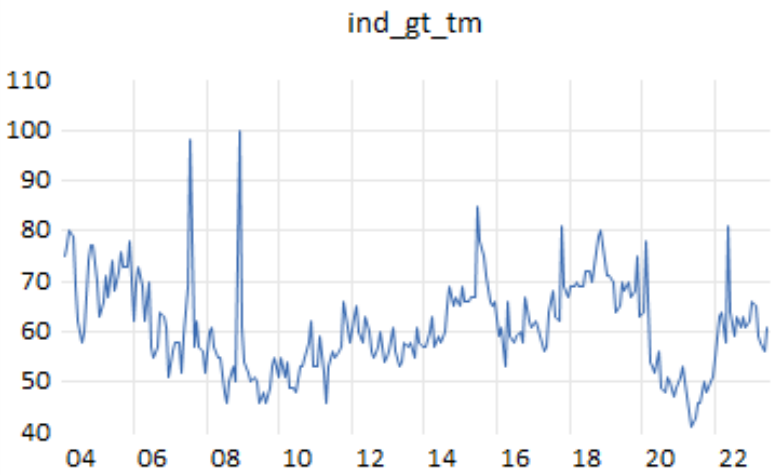
Note: The normalized Google Trend index depends on the share of the Google queries with key words in that specific region on the total number of searches

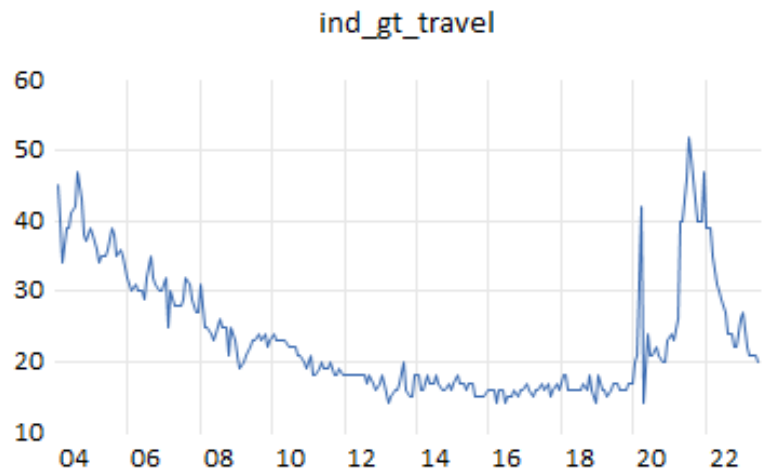
Step 2. Visualize Google Trend Data
India

- **ind_gt_flight:** Flight to India
- **ind_gt_hotel:** India Hotel
- **ind_gt_tm:** Taj Mahal
- **ind_gt_travel**: India Travel









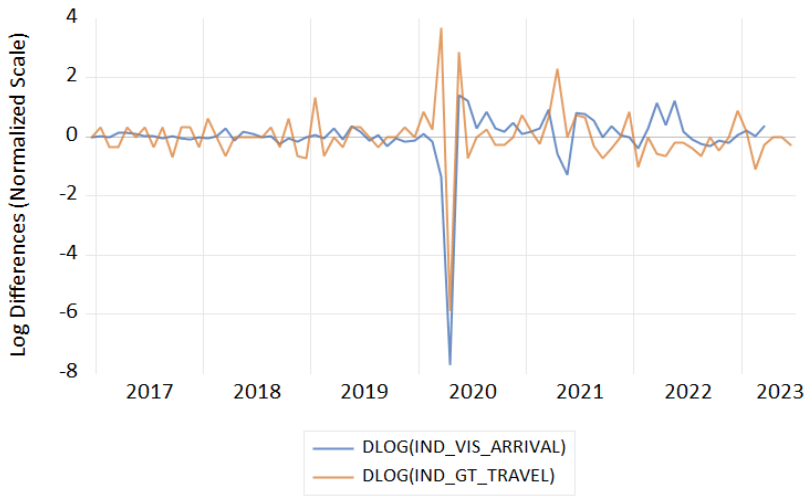
Note: The normalized Google Trend index depends on the share of the Google queries relevant to India tourism worldwide on the total number of searches

Step 2. Visualize Google Trend Data

India: Correlation Matrix

Covariance Analysis: Ordinary
Date: 06/18/23 Time: 05:58
Sample: 2005M02 2023M03
Included observations: 218
Balanced sample (listwise missing value deletion)

Covariance Correlation	DLOG(IND_VIS_ARRIVAL)	DLOG(IND_GT_TM)	DLOG(IND_GT_FLIGHT)	DLOG(IND_GT_HOTEL)	DLOG(IND_GT_TRAVEL)
DLOG(IND_VIS_ARRIVAL)	0.124 1.000				
DLOG(IND_GT_TM)	0.003 0.080	0.013 1.000			
DLOG(IND_GT_FLIGHT)	0.017 0.408	0.000 -0.018	0.013 1.000		
DLOG(IND_GT_HOTEL)	0.010 0.269	0.003 0.286	0.002 0.178	0.010 1.000	
DLOG(IND_GT_TRAVEL)	0.022 0.489	0.000 -0.019	0.007 0.496	0.004 0.293	0.016 1.000

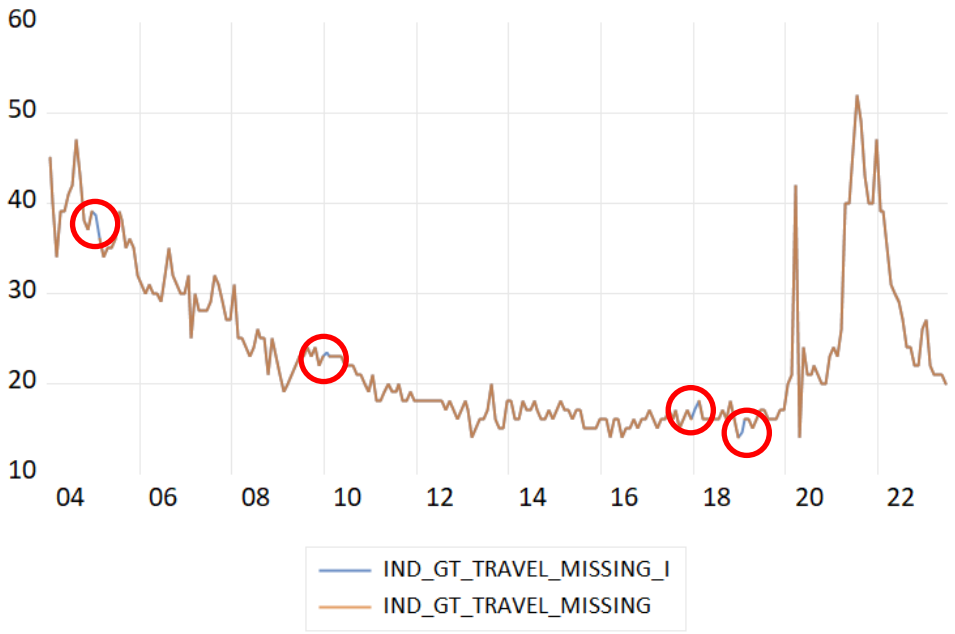


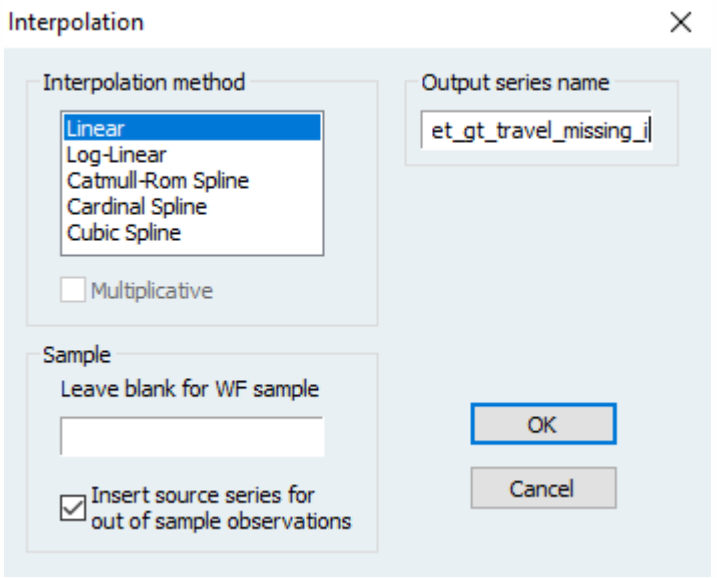
Step 3. Apply Necessary Data Processing

1) Fill-in Missing Data: Interpolation, drop-down menu

Since there's no missing value in Google Trends series, we will create a series with missing value for this practice: remove 2005M01, 2010M01, 2018M01, and 2019M01 data from "**ind_gt_travel**" to create "**ind_gt_travel_missing**" in workfile "**google_trend_tourism_v1.wf1**".

Then, we apply the interpolation method to fill-in the missing values (Proc => Interpolation)





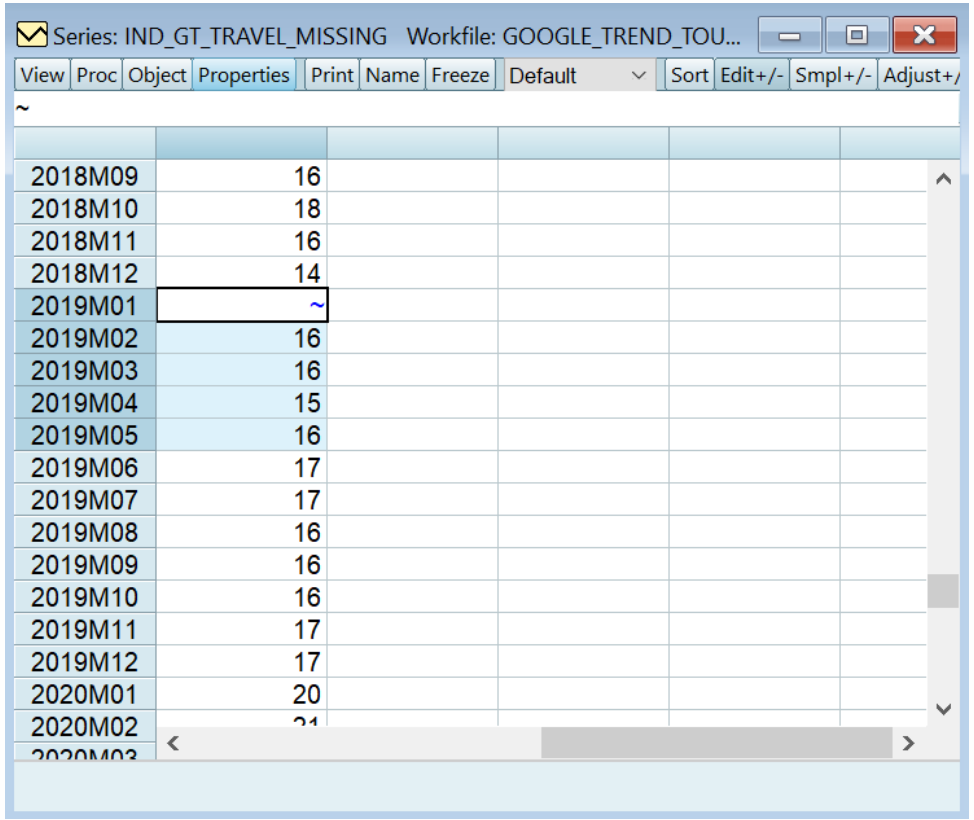
Source: EViews 12 Users Guide I, Chapter 11. Series - Interpolation

- **Linear**: simply computes a linear approximation
- **Log-Linear**: calculated the same way as the Linear method, but the series values are logged before the calculation
- **Cardinal Spline**: based on the previous two non-missing values, the next to non-missing values, and tries to fit the missing data to a non-linear pattern
- **Catmull-Rom Spline**: special case of the cardinal spline, with the tension parameter set to 0
- **Cubic Spline**: special case of spline interpolation and is used often to avoid range's phenomenon.

Step 3. Apply Necessary Data Processing

1) Fill-in Missing Data: Interpolation, spreadsheet view

Besides the user interface, users may also interpolate missing value (NA) under the edit mode of the Spreadsheet view. The procedure involves first selecting the missing value cells, with values before and after the missing values. Then, type-in the method symbol of the preferred interpolation method in value entry section. Click “Enter” to apply the interpolation method.



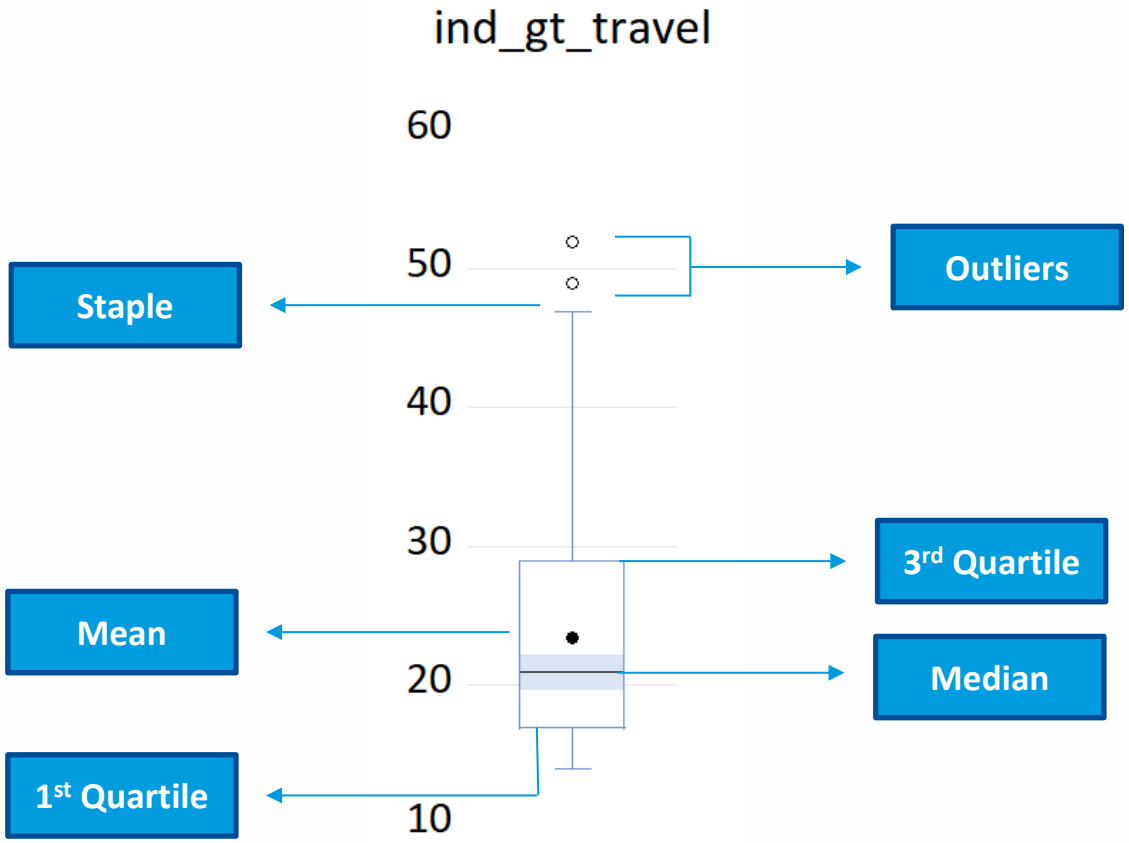
Method symbol	Description
_	Repeats previous non-missing value.
^	Linear interpolation.
~	Cubic spline interpolation.
&	Catmull-Rom spline interpolation.
^*	Log-linear (multiplicative) interpolation (linear in the log of the data).
~*	Multiplicative cubic spline interpolation (a cubic spline on the log of the data).
&*	Multiplicative Catmull-Rom spline interpolation (a Catmull-Rom spline on the log of the data).

Source: EViews 12 Users Guide I, pp 967

Step 3. Apply Necessary Data Processing

2) Remove Outliers: Outliers Detection using Boxplot

The Boxplot, also known as a box and whisker diagram, can be used to detect outliers of a series. The Boxplot summarizes the distribution of a data series by displaying the data using a few primary elements, such as quantiles distribution, staple, mean, median, and outliers.



Step 3. Apply Necessary Data Processing

2) Remove Outliers and Replace using Interpolation

After detecting the outliers, we may replace the outliers using similar method as “fill-in missing values”. In this workshop, we will provide an example to replace outliers with interpolation of the time series. Following EViews codes will be used for the exercise:

```
' sample all
smpl @all

' calculate the mean of original series
scalar mean_ind_gt_travel = @mean(ind_gt_travel)

' calculate the z-score
series z_ind_gt_travel = (ind_gt_travel - mean_ind_gt_travel)/@stdev(ind_gt_travel)

' exclude observations that has z-score >= 2.58 (99% percentile)
smpl @all if @abs(z_ind_gt_travel) < 2.58

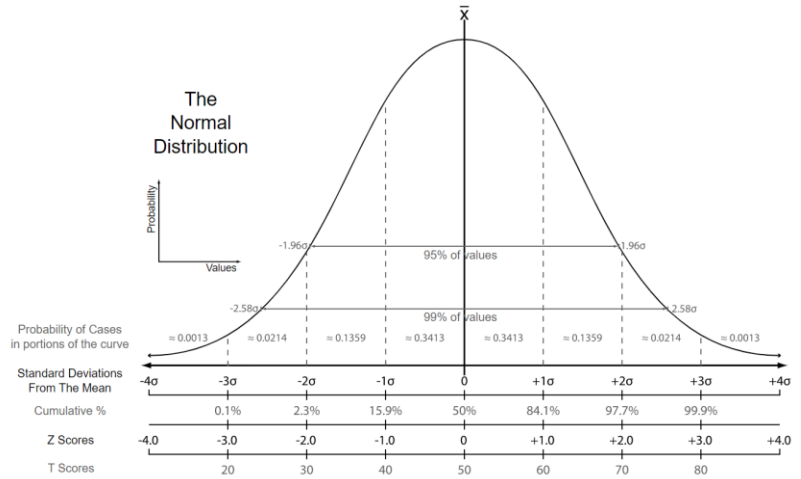
series ind_gt_travel_no_out = ind_gt_travel

' fill-in the outliers with interpolation (Cubic Spline)
smpl @all

ind_gt_travel_no_out.ipolate(type=cb) ind_gt_travel_no_out_i
```

EViews Commands Explain

- **smpl @all if @abs(z_ind_gt_travel) < 2.58**: set Sample size to observations that has smaller than 2.58 **Z-Score**
- **@mean(x)**: calculates average of the values in **x**
- **@abs(x)**: returns absolute value of **x**

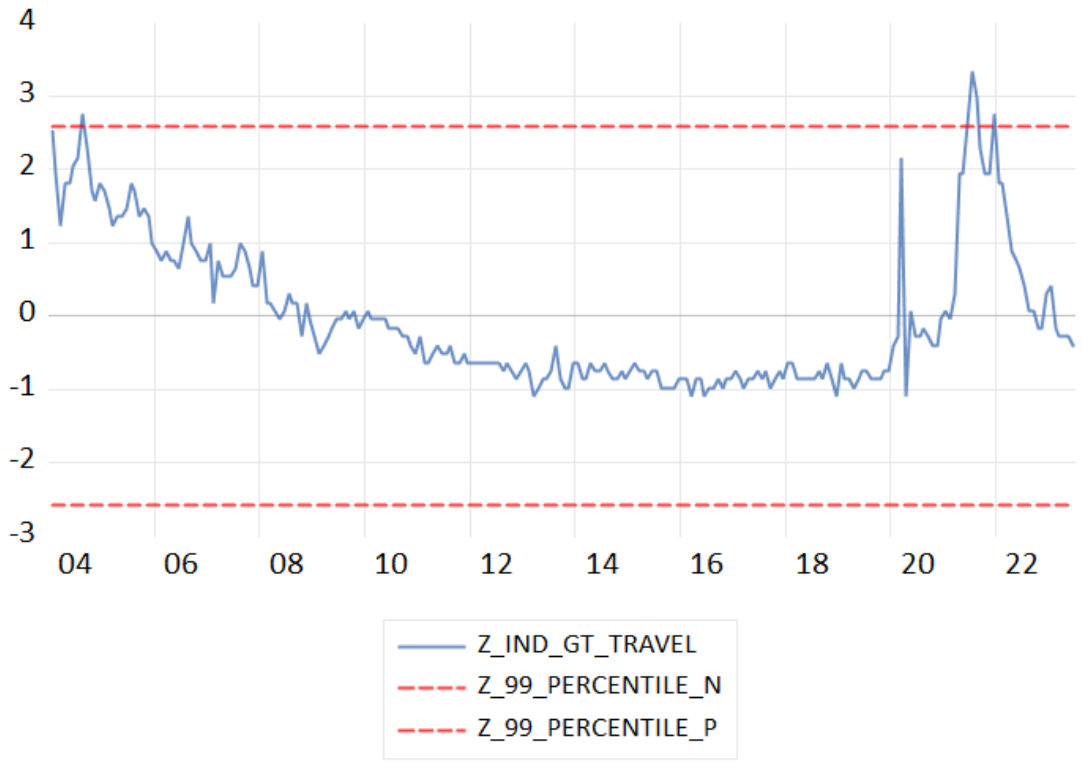


Source: Figure 4.3 on Page 74 of Ward, A. W., Murray-Ward, M. (1999). Assessment in the Classroom. Belmont, CA: Wadsworth. ISBN 0534527043

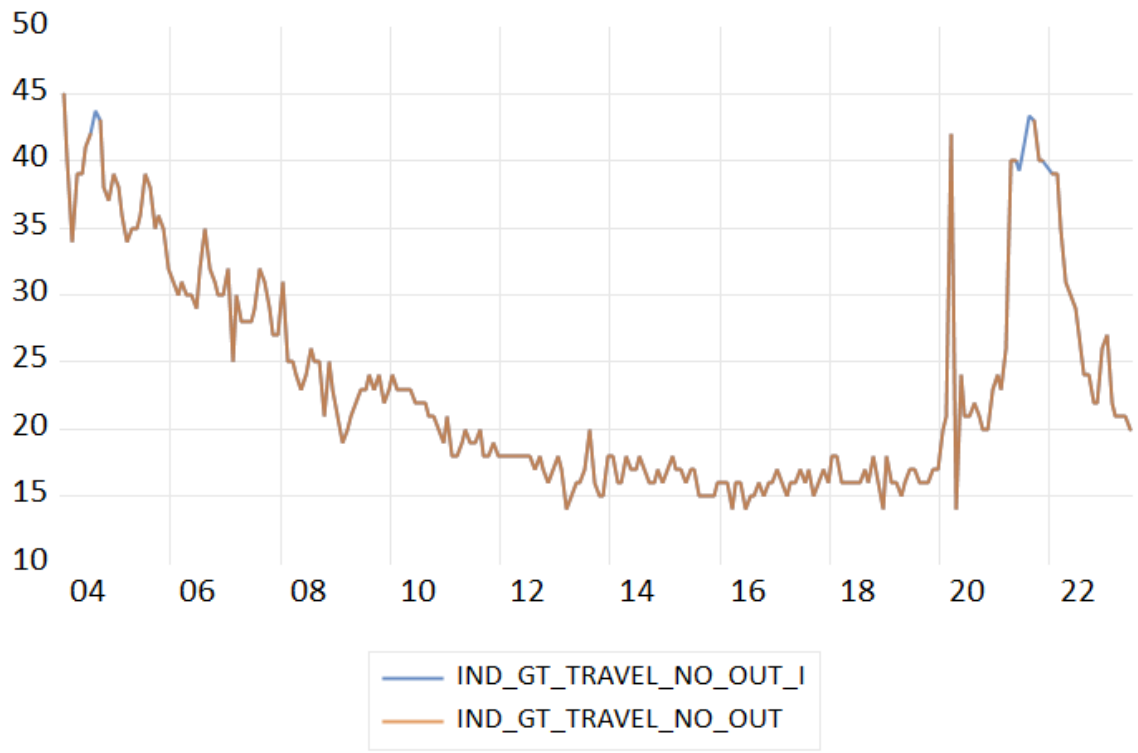
Step 3. Apply Necessary Data Processing

2) Remove Outliers and Replace using Interpolation

Z-Values of IND_GT_TRAVEL



Interpolate Removed Data Points using Interpolation



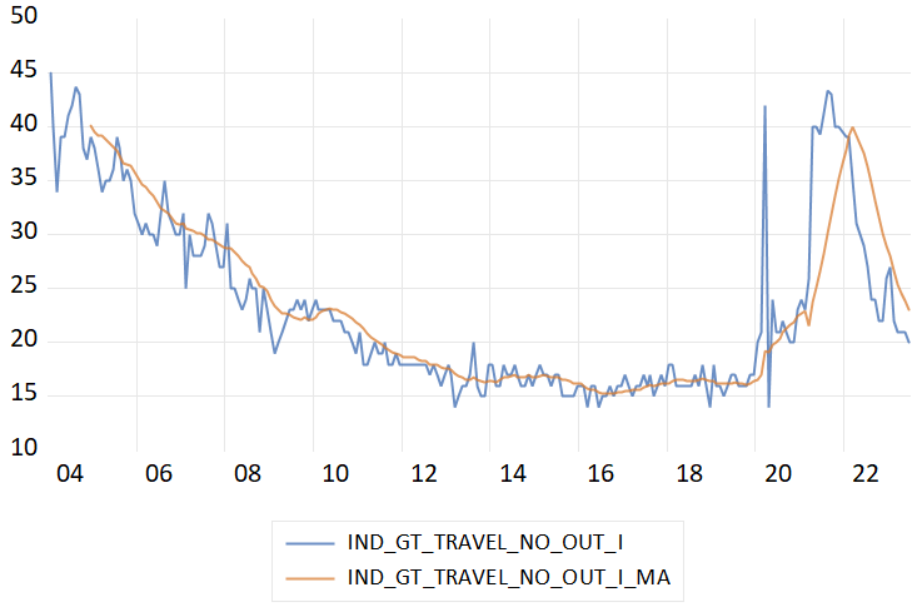
Step 3. Apply Necessary Data Processing

3) Seasonal Adjustment: Smoothing using Moving Averages

Moving averages, the rolling mean, can be used for smoothing out fluctuations and noisy points in a timeseries. We will continue the data processing using the series generated from previous step (“ **ind_gt_travel_no_out_i”**). Following code will be used to create the smoothing series

```
series ind_gt_travel_no_out_i_ma= @movav(ind_gt_travel_no_out_i,12)
```

@movav(x,n) command generates **n**-period backward moving average of series **x**

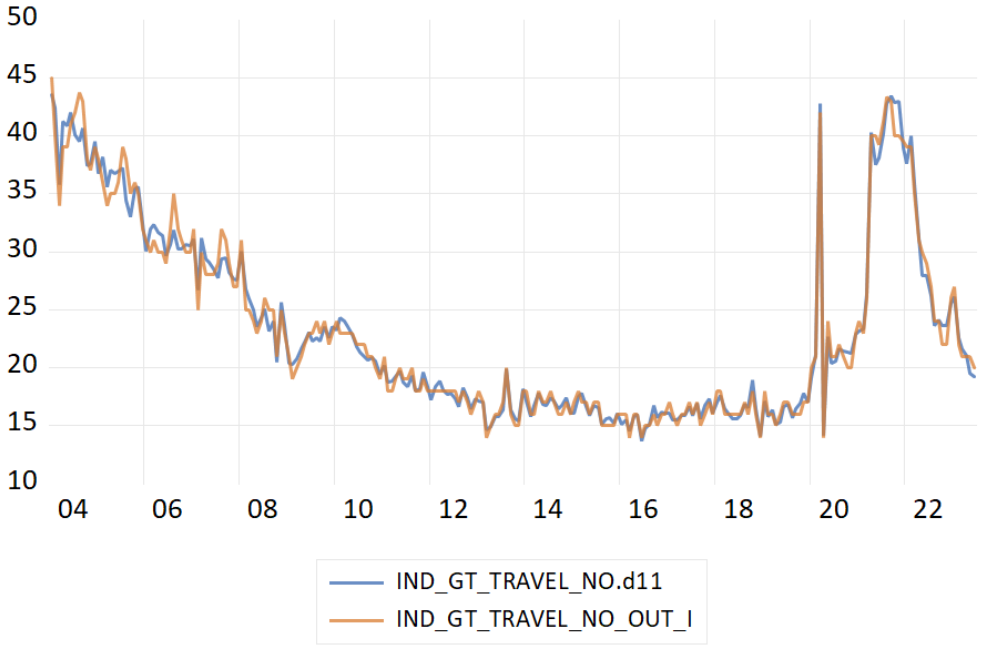


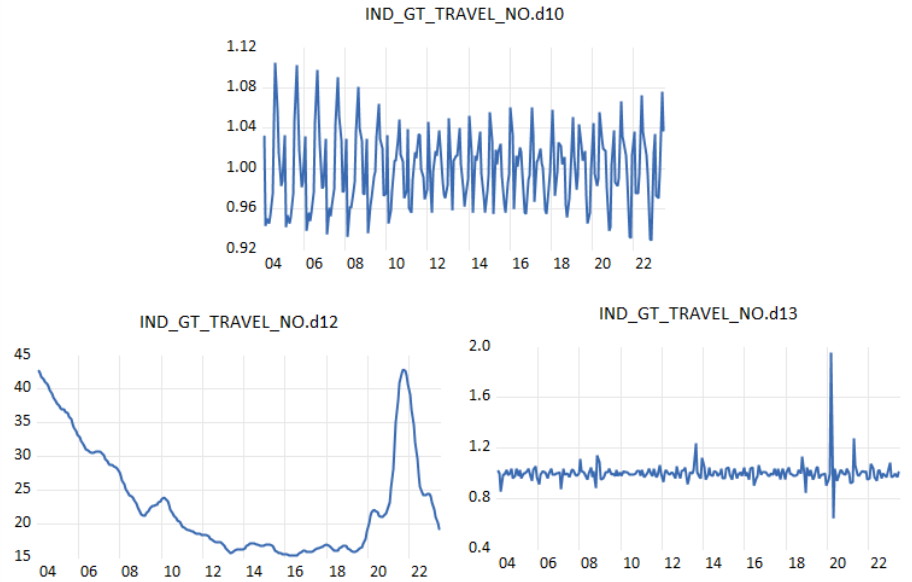
Step 3. Apply Necessary Data Processing

3) Seasonal Adjustment: Decomposition using X-13 Procedure

Since most big data are collected in high frequency, it is very likely that they contain seasonal or other very short run periodicities. Therefore, we need to apply seasonal adjustment to separate the trend and seasonal component from the time series. In this workshop, we will apply the technique learned from Lecture 2 – Part 2 Seasonal Adjustment (Proc => Seasonal Adjustment => Census X-13).

We will also use the series generated after removing outliers ("**ind_gt_travel_no_out_i**").





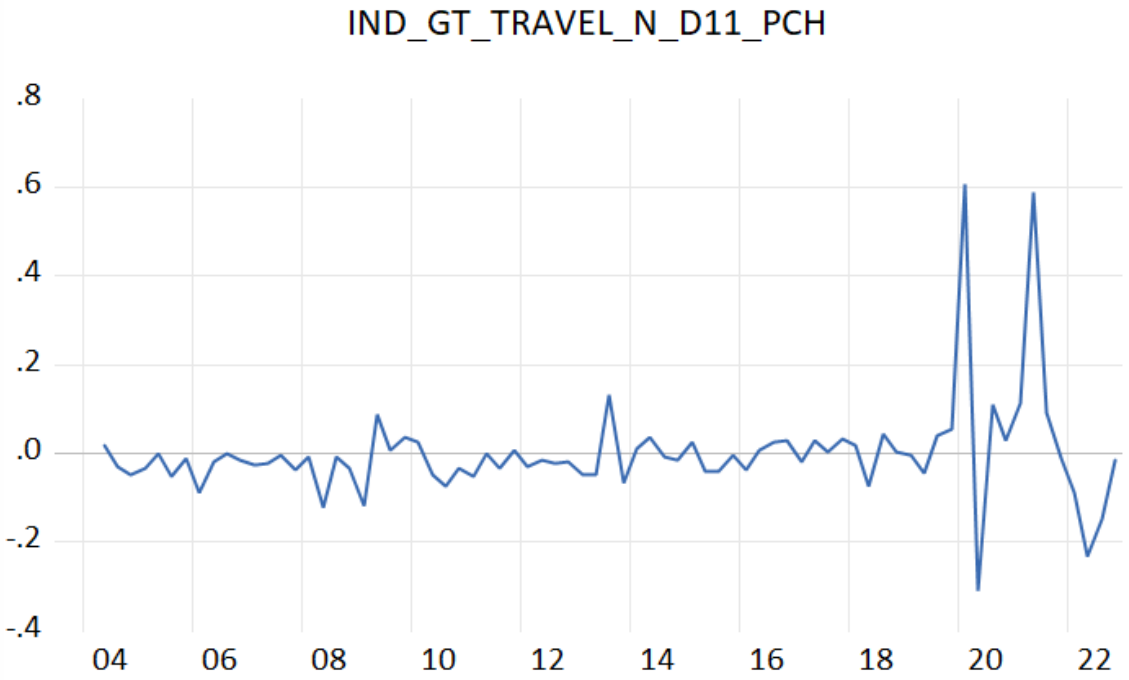
Step 3. Apply Necessary Data Processing

4) Data Transformation: Growth Rate

If we concern the percentage change of a specific variable within a specific time period, we can convert indicators to growth rate. Following code will be used to create the growth rate series (under Monthly sheet):

```
genr ind_gt_travel_n_d11_pch = @pch(ind_gt_travel_n_d11)
```

@pch(x) command will create one-period percentage change (in decimal)



Function	Description
@pc(rc)	calculates the one-period percent change in **rc** (in percent)
@pch(rc)	calculates the one-period percent change in **rc** (in decimal)
@pca(rc)	calculates the one-period ***annualized*** percent change in **rc** (in percent) $(1+@pch(x))^n-1$ (where n=4 for quarterly data, n=12 for monthly, ect)
@pcha(gdp)	calculates the one-period ***annualized*** percent change in GDP (in decimal)
@pcy(gdp)	calculates the one-year percent change (in percent)
@pchy(gdp)	calculates the one-year percent change (in decimal)

Reference

- Google. (2022). Google Trends Help. Retrieved from Google Trends: https://support.google.com/trends/?hl=en#topic=6248052

- Vosen, Simeon; Schmidt, Torsten (January 13, 2011). "Forecasting private consumption: survey-based indicators vs. Google Trends". *Journal of Forecasting*. **30** (6): 565–578. doi:10.1002/for.1213. ISSN 0277-6693.

- Rogers Simon (2016). What is Google Trends data — and what does it mean?. Retrieved from Google News Lab: https://medium.com/google-news-lab/what-is-google-trends-data-and-what-does-it-mean-b48f07342ee8

Thank You!